

1. Let $X_1, \dots, X_n$ be a sample from the exponential distribution with parameter $2/(\alpha + 1)$, that is, the distribution with density

$$f(x; \alpha) = \begin{cases} \frac{2}{\alpha + 1} e^{-2x/(\alpha+1)}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

a) (2 points.) Find the maximum likelihood estimator $\hat{\alpha}$ of the parameter α . Is the estimator $\hat{\alpha}$ unbiased and consistent?

b) (1 point.) Find the method of moments estimator α_1^* using the first moment. Is the estimator α_1^* unbiased and consistent?

c) (2 points.) Find the method of moments estimator α_2^* using the second moment. Is the estimator α_2^* consistent?

d) (2 points.) Check the asymptotic normality of the estimators α_1^* and α_2^* . Compare the estimators α_1^* and α_2^* in the mean-square sense.

e) (1 point.) Compute the Fisher information $I(\alpha)$. Is the estimator α_1^* R-efficient?

2. Let $X_1, \dots, X_n$ be a sample from the uniform distribution on the interval $[1; \theta + 2]$.

a) (2 points.) Find the maximum likelihood estimator $\hat{\theta}$ of the parameter θ . Is the estimator $\hat{\theta}$ unbiased and consistent?

b) (1 point.) Find the method of moments estimator θ^* using the first moment. Is the estimator θ^* unbiased and consistent?

c) (1 point.) Check the asymptotic normality of the estimators $\hat{\theta}$ and θ^* . Compare the estimators $\hat{\theta}$ and θ^* in the mean-square sense.

3 (3 points). Suppose a sample is drawn from the following distribution with θ

X_i	-2	-1	1	2
p_i	$\frac{1}{6}$	$\frac{1}{3} - \theta$	$\frac{1}{3} + \theta$	$\frac{1}{6}$
ν_i	4	6	6	4

where p_i are the theoretical probabilities and ν_i are the observed frequencies (counts). Estimate θ using the maximum likelihood method and using the method of moments (at least two different estimators). Compare these estimators with each other.

(*)(1 point) Which of this estimator will be more efficient?